

Technical Assessment

Please be prepared to present this at your onsite interview. You will be required to bring your own tablet/computer to present during the interview.

Thank you for your interest in a position with the Charles River Development (CRD) / Alpha Platform Engineering organization!

As a part of our technical interview process, we would like to ask each candidate to participate in a *technical assessment exercise*.

Please read through the problem below and answer the questions highlighted in yellow

Account ABC holds these securities - IBM, MSFT, ORCL, AAPL, HD

Account- ABC with \$100K in total asset (100% is vested)	Security	target%	current%	target Variance	unit price	Output - Number of shares to buy/sell
	IBM	20	10	-10	150	
	MSFT	20	20	0	90	
	ORCL	20	30	10	220	
	AAPL	20	20	0	450	
	HD	20	20	0	70	

Total Asset - \$100,000

target% indicates what % of total assets, you should hold in your acct

current% shows what % of total assets, you currently have in your Acct

target Variance shows how much deviation you have, (-ve) var means you need to buy (+ve) var means you need to sell

What do you have to do to get to zero target variance?

Assume that this rebalancing problem is your team's application that you are going to test, please write your manual test cases and automated test cases to test the application.

1. Recommended time for this task is 2-4 hours, please feel free to research
2. You can use any language/testing tool to code your automated test cases for this scenario and make any reasonable assumptions
3. The tests should validate that the total number of shares to buy and sell for each security is correct, as this is the application's output
4. Please be prepared to discuss the logic, test cases and coded implementation during the interview.

**If you make any assumptions, please note that